A Further Results

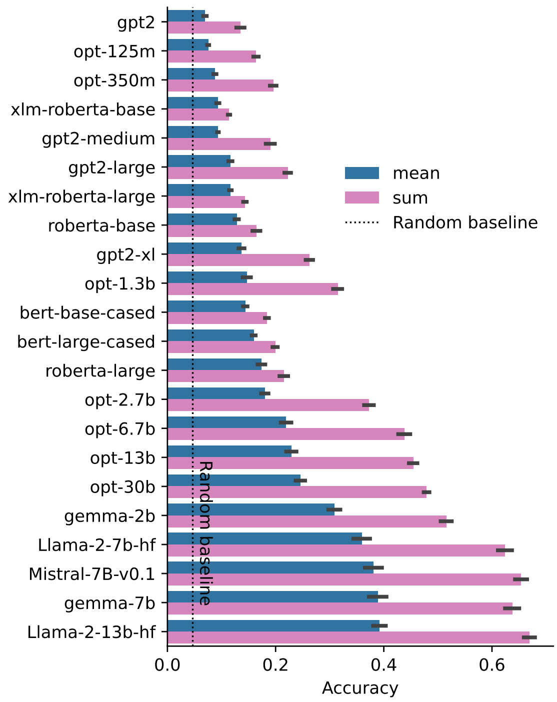


Figure 7: Aggregated accuracy of different retrieval variants on BEAR. The error bars indicate the standard error over three evaluations using the different templates.

A.1 Results on BEAR$_{big}$

In addition to the accuracy on the further refined BEAR dataset, we report the scores on BEAR$_{big}$ (see Table 4). The ranking is very similar and differs only in a few positions.

A.2 Ablation: Pseudo log-likelihood metric

While in preliminary experiments on LAMA, we observed a higher benefit from using the `within_word_l2r` variant, it has only a slightly higher mean score than `original` (see Figure 10). This difference is not significant when using the `sum` variant (p-value of 0.52 on a Student's t-test for paired samples). However, the difference is large when using the `mean` variant (and significant with p-value of 0.025)

A.3 Impact of Pre-Training Data on the BEAR Score

In order to verify the actual ability of the BEAR probe to measure knowledge contained in models' pre-trained weights, we set up an ablation study. We hypothesize that models trained solely on domain-specific datasets, without exposure to the general knowledge tested by BEAR, will show significantly reduced performance compared to those trained on more general datasets, given that the models have similar architectures and sizes. A family of BioGPT models (Luo et al., 2022) was based on the GPT-2 architecture with the sole differences arising from the pre-training data: specifically, they were trained on PubMed abstracts and titles rather than on data from web crawls as it was the case for the GPT models. The evaluation results confirm our hypothesis, with the BioGPT model achieving an average score of 10.84% and its large variant reaching 13.6% on BEAR, both trailing behind the `gpt2-medium` (19.0%) and `gpt2-xl` (26.2%) models.

B Noise Levels

Crowdsourced knowledge bases like Wikidata often contain noisy and inaccurate data. To reduce this issue, we developed heuristics to select better-known entities that are more likely to be verified and corrected. Nonetheless, the potential for noise leakage exists. To evaluate the extent of such noise, we performed a noise levels analysis, in which we manually reviewed 100 randomly chosen examples from various relations for both BEAR$_{big}$ and LAMA (T-REx) probes. We cross-validated the accuracy of the information from Wikidata with alternative sources, confirming 96% and 97% of examples for BEAR$_{big}$ and T-REx, respectively[8].

Additionally, we evaluated whether the object of the relation truly represents the only correct answer as our benchmark (expecting a single correct answer) would mark those answers as incorrect. About 11% of BEAR's answers included multiple correct responses within its answer range, significantly lower than T-REx's 35%[9], which uses BERT's vocabulary as its answer space.

Thus, while BEAR$_{big}$ and LAMA exhibit similar noise levels, BEAR$_{big}$ demonstrates higher reliability by effectively reducing the incidence of multiple correct answers. Moreover, the refined subset of BEAR, due to additional filtering steps, is expected to decrease these values even further.

C GPU Compute Time

Due to the diversity of GPU hardware utilized in our experiments, we have chosen not to present cumulative GPU hours. Instead, to provide insights into the computational requirements of our

[8] A two-sided z-test yields a p-value of 0.6698

[9] A two-sided z-test yields a p-value of 0.0001

Model	Type	# params	BEAR	BEAR$_{1:1}$	BEAR$_{1:N}$
Llama-2-13b-hf	CLM	13b	42.0% ± 0.6%	54.3% ± 1.3%	41.2% ± 0.6%
Mistral-7B-v0.1	CLM	7.0b	41.0% ± 0.6%	52.6% ± 1.2%	40.2% ± 0.6%
gemma-7b	CLM	7.0b	39.5% ± 0.8%	52.0% ± 1.1%	38.6% ± 0.8%
Llama-2-7b-hf	CLM	7.0b	37.5% ± 0.8%	50.3% ± 1.1%	36.6% ± 0.7%
gemma-2b	CLM	2.0b	29.1% ± 0.6%	41.4% ± 1.1%	28.2% ± 0.6%
opt-30b	CLM	30b	25.6% ± 0.3%	35.7% ± 0.9%	24.9% ± 0.3%
opt-13b	CLM	13b	24.2% ± 0.5%	33.6% ± 1.6%	23.5% ± 0.5%
opt-6.7b	CLM	6.7b	23.2% ± 0.7%	33.1% ± 0.5%	22.5% ± 0.7%
opt-2.7b	CLM	2.7b	19.3% ± 0.4%	27.6% ± 0.8%	18.7% ± 0.3%
opt-1.3b	CLM	1.3b	16.0% ± 0.4%	23.3% ± 1.0%	15.5% ± 0.4%
gpt2-xl	CLM	1.6b	12.9% ± 0.2%	17.8% ± 1.2%	12.6% ± 0.2%
roberta-large	MLM	355M	11.1% ± 0.4%	17.1% ± 0.8%	10.7% ± 0.4%
gpt2-large	CLM	812M	10.7% ± 0.2%	14.0% ± 1.5%	10.5% ± 0.2%
bert-large-cased	MLM	335M	10.1% ± 0.3%	11.8% ± 0.7%	10.0% ± 0.3%
bert-base-cased	MLM	109M	9.6% ± 0.3%	11.5% ± 1.2%	9.4% ± 0.3%
opt-350m	CLM	350M	9.5% ± 0.2%	13.4% ± 0.8%	9.2% ± 0.2%
gpt2-medium	CLM	355M	9.1% ± 0.3%	11.3% ± 1.9%	8.9% ± 0.2%
roberta-base	MLM	125M	8.4% ± 0.3%	11.8% ± 1.8%	8.1% ± 0.4%
opt-125m	CLM	125M	8.0% ± 0.2%	9.5% ± 0.8%	7.9% ± 0.2%
xlm-roberta-large	MLM	561M	7.4% ± 0.2%	11.2% ± 1.6%	7.1% ± 0.1%
gpt2	CLM	137M	6.4% ± 0.3%	5.8% ± 1.6%	6.5% ± 0.2%
xlm-roberta-base	MLM	279M	5.8% ± 0.1%	8.7% ± 0.9%	5.6% ± 0.0%
Random Baseline	-	-	2.5%	0.5%	2.7%

Table 4: Models investigated in this work evaluated on BEAR$_{\text{big}}$ (weighted average over all relations) and as the mean over all templates (with the standard error).

research, we offer a selection of representative examples highlighting individual experiments. The duration for the BEAR probe evaluations notably differed across models. For instance, evaluating the `opt-6.7b` model on the larger variant of the probe using the NVIDIA A100 (80GB) machine required approximately 14 hours, whereas `gpt2` averaged about 1 hour. Additionally, causal language models were evaluated faster due to their method for calculating sentence log-likelihoods. For instance, evaluating the `bert-base-cased` model, roughly the same size as `gpt2`, on the entire BEAR probe took four times as long.

Moreover, as stated in the main text, assessing the large variant of BEAR requires a significant amount of time. For example, `bert-base-cased` requires almost 4.5 hours when evaluating the BEAR$_{\text{big}}$ probe with NVIDIA RTX 3090. Yet, this duration drops to around 16 minutes when using the smaller BEAR probe on the same hardware.

D Predictions' Confidence

Log-likelihoods assigned to textual statements by language models over a closed set of answers can be converted into values that can be interpreted as *probabilities*. Here, the idea involves transforming these negative values with the softmax function to obtain a normalized set of values that together resemble a probability distribution. These scores reflect the model's confidence in each possible answer within the set. Furthermore, by comparing the calculated probability distribution with a uniform distribution, we can derive a *uncertainty score*, which reflects the model's confusion for a given instance. This score is based on the Kullback-Leibler Divergence between the predicted probabilities and the uniform distribution. Specifically, we derive divergence between a predicted probability distribution P and a uniform distribution U over N possible answers (outcomes). We then normalize this score to make it scale-free and bounded between 0 and 1 (entropy is maximal for the uniform distribution). Finally, we take a complement of the obtained value (by subtracting it from 1) effectively reversing its interpretation, shifting the focus from divergence to similarity to the uniform distribution. The mathematical formulation of this metric can be found in Equations 1, 2, 3.

$$\begin{aligned} KL(P\|U) &== \sum_{i=0}^{N} p_i \log \frac{p_i}{1/N} \\ &= \sum_{i=0}^{N} p_i \log p_i + \sum_{i=0}^{N} p_i \log N \\ &= \log N - H(P) \\ &= H(U) - H(P) \\ &\in [0, H(U)] \end{aligned} \tag{1}$$

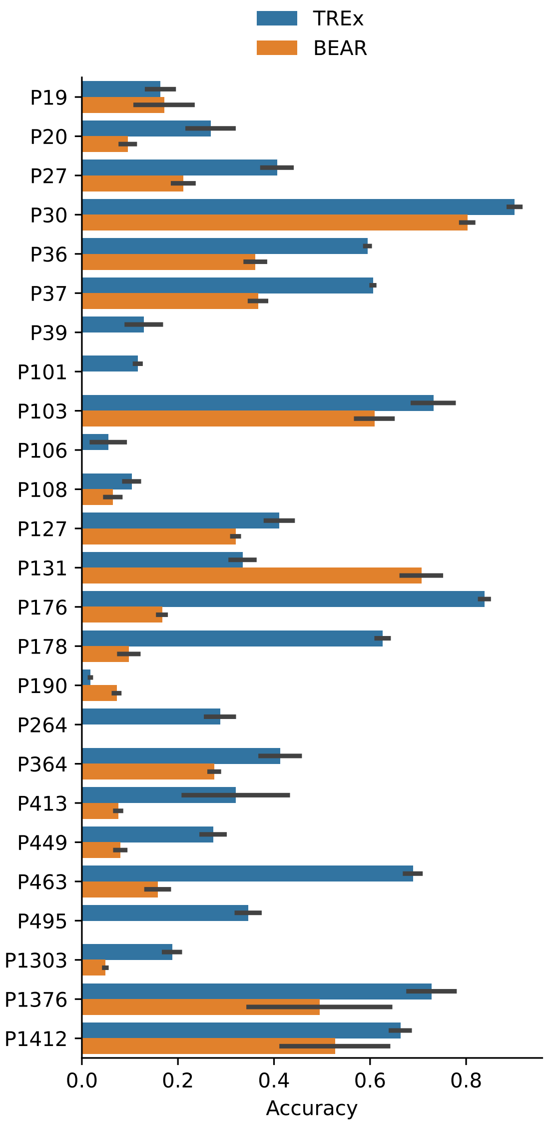
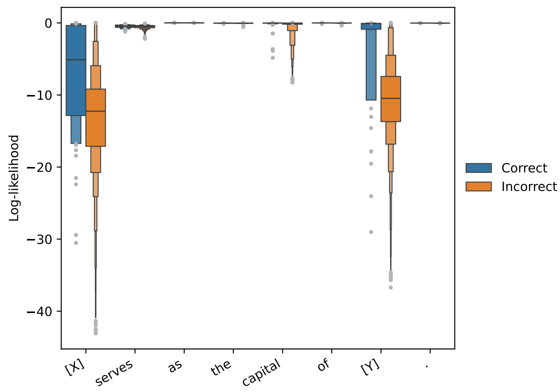
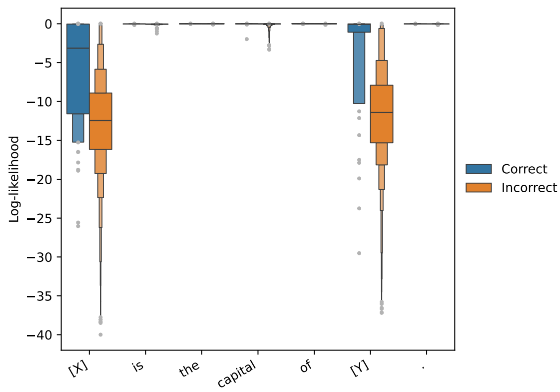


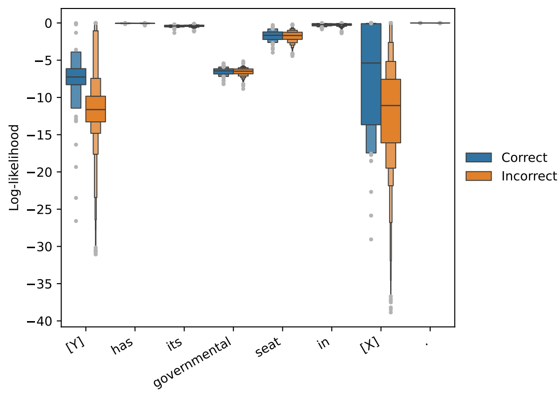
Figure 8: Performance of `bert-base-cased` on a per-relation basis for both BEAR and T-REx probes. The results were obtained by summing pseudo log-likelihood scores (`within_word_l2r`)



(a) First Template: "[X] is the capital of [Y]."; Accuracy of 63%



(b) Second Template: "[Y] has its governmental seat in [X]."; Accuracy of 20%



(c) Third Template: "[X] serves as the capital of [Y]."; Accuracy of 65%

Figure 9: `bert-base-cased` on P1376 (BEAR)

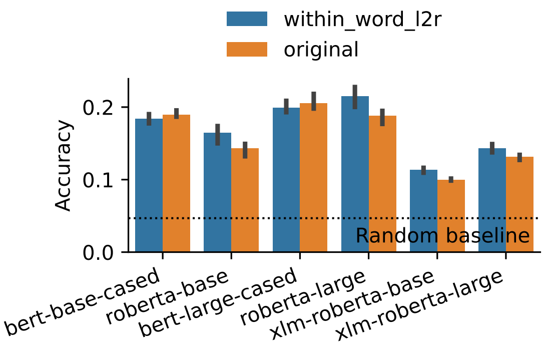


Figure 10: Aggregated accuracy of different retrieval variants on BEAR. The error bars indicate the standard error over three evaluations using the different templates.

$$\text{Normalized KL} = 1 - \frac{H(P)}{H(U)} \in [0, 1] \quad (2)$$

$$\begin{aligned}\text{Uncertainty} &= 1 - \text{Normalized KL} \\ &= \frac{H(P)}{H(U)} \in [0, 1].\end{aligned} \quad (3)$$

The interpretation of such a score is as follows: we observe the largest uncertainty when the predicted distribution approximates a uniform distribution, and the lowest when the entropy of the predicted distribution is null. The latter situation occurs with a point mass distribution, which allocates all the probability mass to one outcome.

E Error Analysis

We further conducted an error analysis on the mispredictions generated by the `Llama-2-13b-hf` model. The investigation did not reveal the presence of any systematic categories or clusters of errors. Some errors appeared to be similar to educated guesses. For instance, `Llama-2-13b-hf` outputs the highest log-likelihood for the factually incorrect statement such as “Kazım Ayvaz passed away in Turkey”. However, this is not an unreasonable assumption, given that *Kazım Ayvaz* was a Turkish Olympic medalist born in Turkey. Even though the correct answer is *Sweden*, this illustrates that some of the model’s errors may stem from logical assumptions instead of arbitrary mistakes.

On the other hand, certain errors appear to be random, with the predicted answer having no apparent connection to the subject. For example, `Llama-2-13b-hf` assigned the highest score to the incorrect answer of *Massif Central* (highlands in France) when queried about the location of the Welsh mountain *Elidir Fawr*.

Furthermore, on rare occasions, the accuracy of the model’s predictions is compromised by issues such as noise leakage or the imperfect quality of data sourced from Wikidata (for the analysis of such noise, see Section B in the Appendix). For instance, a Wikidata entry mentioning *Ronaldo* refers to Ronaldo Luís Nazário de Lima, commonly known simply as Ronaldo, a well-known player for the Brazilian national football team. However, the model erroneously identifies *Ronaldo* as Cristiano Ronaldo, a renowned Portuguese footballer. Table 5 presents a selection of errors made by the `Llama-2-13b-hf` model and provides further information on prediction’s confidence as described in Section D. Specifically, it gives the ranks of the correct and predicted statements as assessed by their log-likelihoods as well as their probability scores obtained by applying the softmax function over all log-likelihoods in the rankings.

F Prompts Used

All prompts were passed as ‘system messages‘ to Chat-GPT4 API and are presented in Figures 11, 12, and 13.